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Name Brandon Buchanan Quiz 6 2/1/2007

1. Write a truth table for (p^q) v~q.

p	q	$(p \wedge q)$	$\sim q$	Q
F	F	F	T	T
F	T	F	F	F
T	F	F	T	T
T	T	T	F	T

2. P(x) : "the word x contains the letter c"
   a) Give a x where P(x) is false

   x = dinosaur

3. Which are propositions
   a) Do not pass go. no
   b) What time is it? no
   c) There are no black flies in Maine. yes
   d) 4+x=5 no
   e) x+1=5, if x=1 yes

4. C(x): x has a cat. D(x): x has a dog F(x): x has a ferret. Express the statement. Assume x can only be students in Discrete.
   a) A student in Discrete has a cat, a dog, and a ferret. $C(x) \wedge D(x) \wedge F(x)$
   b) All students in your class have a cat, a dog, and a ferret. $\forall x \; C(x) \wedge D(x) \wedge F(x)$
   c) Some student in Discrete has a cat and a dog, but not a ferret. $\exists(x) \; C(x) \wedge D(x) \wedge F(x)$
   d) No student in Discrete has a cat, a dog, and a ferret. $\sim\forall x \; C(x) \wedge D(x) \wedge F(x)$

5. Negate the statement: 3 is odd and today is not Monday

   3 is not odd or today is Monday.

6. Is $\forall x \exists y$ P(x,y) true or false, when P(x,y):xy=1 false (just taking a guess)
7. Is $\exists x \forall y$ P(x,y) true or false, when P(x,y):xy=1 false